



CONSERVATIVE INTEGRATORS FOR PIECEWISE SMOOTH SYSTEMS WITH TRANSVERSAL DYNAMICS

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ABSTRACT. We introduce conservative integrators for long term integration of piecewise smooth systems with transversal dynamics and piecewise smooth conserved quantities. In essence, for a piecewise dynamical system with piecewise defined conserved quantities where its trajectories cross transversally to its interface, we combine Mannshardt's transition scheme and the Discrete Multiplier Method to obtain conservative integrators capable of preserving conserved quantities up to machine precision and accuracy order. We prove that the order of accuracy of the conservative integrators is preserved after crossing the interface in the case of codimension one number of conserved quantities. Numerical examples in two and three dimensions illustrate the preservation of accuracy order across the interface for cubic and logarithmic type conserved quantities. We observed that conservative transition schemes can prevent spurious transitions from occurring, even in the case when there are fewer conserved quantities.

1. Introduction. Piecewise defined ordinary differential equations (ODEs) arise naturally in a variety of fields including electrical circuits, mechanical systems, biology and others. Electrical circuits with an ideal diode have different dynamics based on whether the diode is conducting electricity or not [1]. In mechanical systems with friction, the Coulomb model of friction allows for a sudden change in the frictional force depending on whether the object is moving or stationary. In gene regulatory networks the dynamics depends on which genes are turned on or off [19].

We develop a conservative numerical method for integrating ODEs corresponding to piecewise defined vector fields. We refer to such systems of ODEs as piecewise smooth systems (PWS). Our results depend only on the local piecewise structure of the vector fields. Locally, we assume the phase space is subdivided into two

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subdomains separated from each other by a smooth codimension 1 embedded submanifold called the *switching surface*, or equivalently the *interface*. For example, in two dimensions, the phase space consists of two subdomains separated by a simple smooth curve and in three dimensions, the two subdomains are separated by a smooth surface. We do not address the case of higher codimension switching submanifolds such as a curve singularity in three dimensions [11].

A piecewise defined vector field is defined on both subdomains with a different definition on each side of the interface. The vector field is not required to be defined on the switching surface. However, in our subsequent analysis, we require that the vector fields from the two sides be continuous up to the interface, without requiring their limit from both sides agree on the interface. Moreover, we will require the vector field to be transverse to the surface and point in such a way as to allow the piecewise trajectories to continue after arriving at the surface. Thus, we do not consider sliding motion on the switching surface [19]. Furthermore, we assume the smooth vector fields defined on each subdomain satisfy the standard conditions needed for local existence and uniqueness of solutions.

Our main goal is long term integration of such vector fields across repeated crossings of the switching surface, while preserving one or more conserved quantities, regardless of whether the conserved values or conserved quantities may be different in each of the two subdomains.

The theory of ODEs with discontinuous right hand sides is well-developed. Filippov and others [7] have generalized the definition of solution of such an ODE by replacing the equation with a differential inclusion (so that at the discontinuities the vector field belongs to a set rather than being equal to a specific value) and investigated conditions for existence and uniqueness and dependence on parameters and initial conditions. The appearance of chaos and bifurcation theory for such systems has been another direction of investigation [3]. Numerical methods for such ODEs have also been developed [17, 8, 2]. For a survey of such methods see [4].

Geometric numerical methods have also been applied to more complex PWS, such as contact dynamics involving rigid or elastic bodies [5, 6, 10, 13, 14, 15]. In [6], nonsmooth Lagrangian mechanics is discretized using a discrete variational principle leading to geometric variational integrators for rigid body motion with collisions. This approach works when a Lagrangian formulation is available and discrete conservation of the energy, momentum or symplectic form is sought. In [13], Verlet integrators were combined with kinetic data structures and discretization of contact barrier potential to achieve impressive nonsmooth mechanics simulations with an emphasis on the algorithms and data structures for organizing the asynchrony. In [14], discrete gradient integrators have been used on an active set of contact nodes and in [5, 10], heterogeneous asynchronous time integrators have also been employed to provide discrete energy balance. Finally in [15], an algorithm for nonsmooth contact geometry is developed based on the use of nonsmooth analysis and quadratic programming. In contrast, our work focuses on integrating simpler PWS systems, with a codimension one interface surface, that exactly preserve conserved quantities, as well as studying how geometric information can improve accuracy.

The main contribution of this paper is the development of a *conservative* numerical method with accuracy guarantees, for piecewise smooth dynamics in which there are possibly different conservation laws before and after a trajectory encounters the switching surface. We do this by extending the *Discrete Multiplier Method* (DMM) [22] for dynamical systems with conserved quantities to yield conservative

numerical schemes which allow for long term integration with repeated interface crossings and jumps in the conserved quantity across the switching surface. To the best knowledge of the authors, our approach is the first numerical method for PWS systems that can exactly preserve multiple conserved quantities of arbitrary forms over long term integration. The effectiveness of the numerical method is demonstrated on two conservative systems of nonlinear ODEs with discontinuous right hand sides. For a d -dimensional phase space with a time-independent interface, when there are $d - 1$ number of conserved quantities on the smooth parts of the phase space, we obtain a different proof than Mannshardt's original work on the preservation of accuracy order when crossing the interface. Specifically, this stronger geometric hypothesis allows us to prove the transition time can be automatically computed accurately, in contrast to Mannshardt's original work on order preservation which required the transition time to be computed sufficiently accurately a priori. Moreover, numerical experiments show combining transition scheme with conservative integrators can prevent spurious transitions when compared to using traditional integrators, even in the case when there are fewer conserved quantities.

2. Background. For d, d', k positive integers, $U \subset \mathbb{R}^d$ and $V \subset \mathbb{R}^{d'}$ open subsets, $C(U \rightarrow V)$ is the space of continuous functions and $C^k(U \rightarrow V)$ is the space of functions all of whose partial derivatives up to k -th order exist and are continuous. A closed ball of radius $r > 0$ with a center at $\mathbf{a} \in \mathbb{R}^d$ will be denoted $B_r(\mathbf{a})$. The setting we are interested in is a PWS system on a bounded open interval I with its phase space on a bounded open subset $U \subset \mathbb{R}^d$ separated into two parts by a codimension 1 hypersurface. Let $g \in C^1(U \rightarrow \mathbb{R})$ with $\nabla g \neq \mathbf{0}$ on U (called a switching function) define a switching surface $S := \{\mathbf{x} \in U \mid g(\mathbf{x}) = 0\}$, which is a C^1 $(d - 1)$ -dimensional embedded submanifold of $U \subset \mathbb{R}^d$. We denote the two parts of the phase space as $U_{\pm}$, where $U_+ := \{\mathbf{x} \in U \mid g(\mathbf{x}) > 0\}$ and $U_- := \{\mathbf{x} \in U \mid g(\mathbf{x}) < 0\}$, and assume $U_{\pm} \neq \emptyset$ ¹. Assume existence of $\mathbf{f}_{\pm} \in C^1(I \times (U_{\pm} \cup S) \rightarrow \mathbb{R}^d)$ defining a PWS system:

$$\dot{\mathbf{x}}(t) = \begin{cases} \mathbf{f}_-(t, \mathbf{x}(t)), & \text{if } \mathbf{x}(t) \in U_-, \\ \mathbf{f}_+(t, \mathbf{x}(t)), & \text{if } \mathbf{x}(t) \in U_+, \end{cases} \quad (1)$$

$$\mathbf{x}(t_0) = \mathbf{x}_0 \in U \setminus S.$$

As $\dot{\mathbf{x}}(t^*)$ is undefined on $\mathbf{x}^* = \mathbf{x}(t^*) \in S$, we need to decide on the dynamics upon reaching the interface S . We shall assume that the trajectories are transversal to the switching surface and are such that the trajectories approaching the switching surface from one subdomain will exit into the other subdomain. This condition, known as the *transversality condition* is satisfied if there exists a nonzero $\alpha_S \in \mathbb{R}$, referred to as the *transversality constant*², so that,

$$(\nabla g \cdot \mathbf{f}_{\pm})(t^*, \mathbf{x}^*) \geq \alpha_S^2, \text{ for any } (t^*, \mathbf{x}^*) \in I \times S. \quad (2)$$

(We have defined transversality as both $(\pm)$ terms $(\nabla g \cdot \mathbf{f}_{\pm})$ being positive. To accommodate both being negative one can redefine the switching function $\tilde{g} := -g$.)

From Filippov's first order theory [7] on PWS systems, if the transversality condition is satisfied, any trajectory $\mathbf{x}(t)$ of (1) not starting on the interface and reaching

¹If either of $U_{\pm}$ is empty, then the dynamical system is smooth.

²In the notation for the transversality constant, we emphasize the dependence on S but not on time. This is because in the analysis of the key estimates we will be interested in crossing S in a particular time interval.

S at t^* will exit S after t^* and is unique. Moreover, we wish to exclude the possibility of infinitely many transitions across S over a finite time interval. For this reason and for later proofs, throughout this paper, we assume that the solutions $\mathbf{x} \in C(I \rightarrow U)$ of (1) satisfy the following hypotheses:

- (H1) There exist at most finitely many *transition times* $t^* \in I$, $\mathbf{x}(t^*) \in S$.
- (H2) The one-sided limits of $\dot{\mathbf{x}}$ approaching S exist and are equal to $\mathbf{f}_\pm$ from either $U_\pm$. Specifically, there exists a fixed $\epsilon_0 > 0$ such that for $0 < \epsilon < \epsilon_0$,

$$\begin{aligned} \lim_{t \downarrow t^*} \dot{\mathbf{x}}(t) &= \mathbf{f}_\pm(t^*, \mathbf{x}^*), \text{ if } \mathbf{x}(t^* + \epsilon) \in U_\pm, \\ \lim_{t \uparrow t^*} \dot{\mathbf{x}}(t) &= \mathbf{f}_\pm(t^*, \mathbf{x}^*), \text{ if } \mathbf{x}(t^* - \epsilon) \in U_\pm. \end{aligned}$$

- (H3) Denoting the finitely many transition times t_i^* as $a \leq t_1^* < \dots < t_r^* \leq b$ with $[a, b] \subset I$, we require that $\mathbf{x} \in C^2(I \setminus \{t_1^*, \dots, t_r^*\} \rightarrow U)$ and $\|\ddot{\mathbf{x}}(t)\|_2 < \infty$ for $t \in I \setminus \{t_1^*, \dots, t_r^*\}$.

Our integrator that we will describe is in the class of PWS integrators called *event driven* methods. Specifically, the integrator arrives near S using the right hand side in (1), is stopped just prior to S and is restarted using the other right hand side. The original main numerical results are in [17] showing that accuracy after crossing S is limited by how accurately the arrival point on S is computed. Ignoring the discontinuity reduces the accuracy order, for example see [4] for a survey. We will refer to such a method as a *transition scheme*. Our goal is to extend transition schemes to preserve conserved quantities.

A conserved quantity is a function that is constant along the integral curves of the ODE and an ODE may possess several conserved quantities. A basic example is the total energy $\psi(x, v) = \frac{1}{2}(mv^2 + kx^2)$ which is constant along the solutions of the harmonic oscillator $v = \dot{x}, \dot{v} = -\frac{k}{m}x$. Conserved quantities (also referred to as constants of motion) may also exist for dissipative systems. For example, $\psi(t, x, v) = \frac{1}{2}(mv^2 + \gamma xv + kx^2) \exp(\gamma t/m)$ is constant along the solutions of the damped harmonic oscillator $v = \dot{x}, \dot{v} = -\frac{\gamma}{m}v - \frac{k}{m}x$, see [22]. We will be using conservative integrators based on DMM [24, 22, 25]. These integrators preserve arbitrary forms of conserved quantities up to machine precision and have been recently applied to many body problems [23, 9] and Hamiltonian Monte Carlo [18] and geodesics in Schwarzschild geometry [21].

We will only consider a PWS system (1) with solutions satisfying (H1)-(H3) that have time-independent conserved quantities $\psi_\pm \in C^1(U_\pm \cup S \rightarrow \mathbb{R}^{d_\pm})$, which are linearly independent³.

3. Conservative integrator for PWS systems. In this section, after introducing some notations, we will introduce the conservative integrator and discuss various related well-posedness results.

3.1. Notations. Throughout the rest of the paper, τ will refer to the maximum over all discrete time step sizes and $\|\cdot\|$ denotes the ℓ_2 norm of vectors. For simplicity and convenience, we will assume that the time step size is fixed although all results apply with minimal changes to the variable time step case. Let T denote the final time and define uniformly distributed time steps $t_k = t_0 + k\tau$, $N = (T - t_0)/\tau$, so that $T = t_0 + N\tau$ and $\tau = (T - t_0)/N$.

³By linear independence, we mean that $\nabla \psi_\pm$ have full row rank on $U_\pm$, see Definition 11 of [22]

In the expressions where τ appears as a superscript, it is intended to denote discrete expressions that appear in the numerical method and not to the specific length of the discrete time step. Thus, for example, $\mathbf{f}_{\pm}^{\tau}(t_k, \mathbf{x}_k, t_{k+1}, \mathbf{x}_{k+1})$ denotes “discrete vector fields” approximating the PWS vector fields $\mathbf{f}_{\pm}(t, \mathbf{x})$. The discrete solution $\mathbf{x}^{\tau}(t; \mathbf{x}_k, t_k)$ denotes the numerical solution at time t , starting with the initial condition $\mathbf{x}_k$ at t_k and this is supposed to approximate the piecewise smooth solution $\mathbf{x}(t; \mathbf{x}_k, t_k)$ starting with the same initial conditions. Often when it is clear from the context, we will shorten these to $\mathbf{x}^{\tau}(t)$ and $\mathbf{x}(t)$, respectively.

By hypothesis (H3), we can assume any solution curve will cross the interface at most finitely many times. We denote an exact transition time and location at which the interface is crossed by starred quantities with $t^* \in I$ and $\mathbf{x}^* \in U$ and the corresponding discrete transition time $\hat{t}^* \in I$ and point $\hat{\mathbf{x}}^* \in U$. Specifically,

$$\mathbf{x}^* = \mathbf{x}(t^*; \mathbf{x}_k, t_k), \quad (3)$$

$$\hat{\mathbf{x}}^* = \mathbf{x}^{\tau}(\hat{t}^*; \mathbf{x}_k, t_k). \quad (4)$$

3.2. Conservative transition scheme. We will consider single-step conservative numerical schemes for the PWS system (1) of the form

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \tau \mathbf{f}_{\pm}^{\tau}(t_k, \mathbf{x}_k, t_{k+1}, \mathbf{x}_{k+1}), \quad (5)$$

such that $\psi_{\pm}(\mathbf{x}_{k+1}) = \psi_{\pm}(\mathbf{x}_k)$ where $\psi_{\pm}(\mathbf{x})$, the conserved quantity of (1) is constant along the integral curves of the corresponding smooth parts of (1).

We can consider any number $d_{\pm}$ of conserved quantities $\psi_{\pm} \in C^1(U_{\pm} \rightarrow \mathbb{R}^{d_{\pm}})$. In the ideal case, when there are $d_{\pm} = d - 1$ conserved quantities, the shape of the trajectory is completely determined by the conserved quantities and the conservative transition scheme will only induce an error in time.

Using this geometric property, we will show in Section 4 that the resulting conservative scheme can preserve the order of accuracy on crossing the switching surface with a simplified proof than the work of Mannshardt [17].

In order to preserve the accuracy order, a special treatment for (5) is needed when the discrete solution crosses the interface S , which is detected by evaluating $g(\mathbf{x}_k)$ after each discrete time step. If such a crossing is detected at time step t_{k+1} , then the method backs up to time step t_k and follows the transition scheme introduced next.

Assuming $(t_k, \mathbf{x}_k) \in I \times U_-^4$, find $(\hat{t}^*, \hat{\mathbf{x}}^*)$ by solving for $(t, \mathbf{x})$ in

$$\mathbf{x} = \mathbf{x}_k + (t - t_k) \mathbf{f}_{-}^{\tau}(t_k, \mathbf{x}_k, t, \mathbf{x}), \quad (6)$$

$$g(\mathbf{x}) = 0. \quad (7)$$

The error incurred in the numerical solution $(\hat{t}^*, \hat{\mathbf{x}}^*)$ of the system (6)-(7) plays an important role in the analysis of the order preservation across the interface, detailed in Section 4.

Remark 3.1. Note that we only consider time-independent g and so the interface is fixed. The transition scheme can still work in principle for a time-dependent $g(t, \mathbf{x})$. However, our main result in Section 4 on the preservation of order across the interface is only proved in the time-independent case. Preserving order across a time-dependent interface would require the ability to control the time error between the discrete and continuous trajectories. Unless one can control $\hat{t}^* - t^*$, the error in

⁴Analogous steps can be made if $(t_k, \mathbf{x}_k) \in I \times U_+$ by interchanging $\mathbf{f}_{\pm}^{\tau}$.

the time of intersection with the interface, it is difficult to control $\hat{\mathbf{x}}^* - \mathbf{x}$, the error in the location of the intersection.

Assuming a solution $(\hat{t}^*, \hat{\mathbf{x}}^*)$ is found from (6) and (7), the transition scheme finishes by solving for $\mathbf{x}_{k+1}$ in

$$\mathbf{x}_{k+1} = \hat{\mathbf{x}}^* + (t_{k+1} - \hat{t}^*) \mathbf{f}_+^\tau(\hat{t}^*, \hat{\mathbf{x}}^*, t_{k+1}, \mathbf{x}_{k+1}). \quad (8)$$

We summarize Mannshardt's transition scheme in Algorithm 1 below.

Algorithm 1: Mannshardt's transition scheme (uniform time steps)

Given $\tau, t_0, \mathbf{x}_0$.

for $k = 0, 1, 2, \dots$ **do**

$t_{k+1} = t_k + \tau$

$s \leftarrow \text{sign}(g(\mathbf{x}_k))$

$\tilde{\mathbf{x}}_{k+1} \leftarrow \text{Solve for } \mathbf{x} \text{ in } \boxed{\mathbf{x} = \mathbf{x}_k + \tau \mathbf{f}_s^\tau(t_k, \mathbf{x}_k, t_{k+1}, \mathbf{x})}$

$s' \leftarrow \text{sign}(g(\tilde{\mathbf{x}}_{k+1}))$

if $s = s'$ **then**

$\mathbf{x}_{k+1} \leftarrow \tilde{\mathbf{x}}_{k+1}$

else

$(\hat{t}^*, \hat{\mathbf{x}}^*) \leftarrow \text{Solve for } (t, \mathbf{x}) \text{ in } \boxed{\begin{cases} \mathbf{x} = \mathbf{x}_k + (t - t_k) \mathbf{f}_s^\tau(t_k, \mathbf{x}_k, t, \mathbf{x}) \\ g(\mathbf{x}) = 0 \end{cases}}$

$\mathbf{x}_{k+1} \leftarrow \text{Solve for } \mathbf{x} \text{ in } \boxed{\mathbf{x} = \hat{\mathbf{x}}^* + (t_{k+1} - \hat{t}^*) \mathbf{f}_{s'}^\tau(\hat{t}^*, \hat{\mathbf{x}}^*, t_{k+1}, \mathbf{x})}$

end

end

The main steps of Algorithm 1 are the solution of the system (6) and (7) for $(\hat{t}^*, \hat{\mathbf{x}}^*)$ followed by computing the update $\mathbf{x}_{k+1}$. Details on existence and uniqueness of solution are discussed in [26, Appendix A], which we briefly summarize next. [26, Lemma A.1] states a sufficient condition for existence of $(t^*, \mathbf{x}^*)$ the exact transition time and point for the piecewise smooth solution $\mathbf{x}(t)$. A partial converse of this requires transversality and is the content of [26, Lemma A.2]. Assuming that $\hat{t}^*$ exists during a single time step $[t_k, t_{k+1}]$ the remainder of [26, Appendix A] is devoted to proving the existence of $(\hat{t}^*, \hat{\mathbf{x}}^*)$ and $\mathbf{x}_{k+1}$. The existence result is in [26, Proposition A.5] which needs two technical Lemmas [26, Lemma A.3] and [26, Lemma A.4]. These lemmas prove the existence of a certain fixed point (which is equivalent to solving (6)) and the Lipschitz continuity of the discrete solution.

4. Order preservation upon crossing the interface. In this section, we will present our main result on the order preservation for Mannshardt's transition scheme with conservative integrators. Specifically, under a relatively strong hypothesis of the existence of $d - 1$ number of conserved quantities, we prove the transition time can be automatically computed accurately resulting in a different proof than Mannshardt's original work on order preservation, where the transition time needed to be computed sufficiently accurately a priori. First, we will state our main result concerning preservation of order. Following a series of lemmas in Sections 4.1 and 4.2, we then prove the main theorem in Section 4.3.

Theorem 4.1 (Main Theorem). *Let the piecewise smooth trajectory $\mathbf{x}(t; \mathbf{y}, s)$ be the solution of (1), $\mathbf{x}^\tau(t; \mathbf{y}, s)$ be the solution of the transition scheme (6)-(8) and $g \in C^2(\bar{U} \rightarrow \mathbb{R})$ denote the switching function. Assume that the discrete vector fields $\mathbf{f}_\pm^\tau \in C(I \times (U_\pm \cup S) \times I \times (U_\pm \cup S) \rightarrow \mathbb{R}^d)$ are Lipschitz continuous in their arguments. Moreover, assume the piecewise smooth trajectory $\mathbf{x}(t; \mathbf{y}, s)$ and the discrete trajectory $\mathbf{x}^\tau(t; \mathbf{y}, s)$ are Lipschitz continuous with respect to their initial data $\mathbf{y}, s$. Furthermore, assume that the transition schemes (6)-(8) are conservative for $d-1$ time-independent conserved quantities $\psi_\pm$ of the piecewise smooth system (1), where $\nabla \psi_\pm$ have full row rank in $U_\pm \cup S$, respectively. If $\hat{t}^* \in [t_k, t_{k+1}]$, then for sufficiently small τ ,*

$$\|\mathbf{x}(t_{k+1}; \mathbf{x}_k, t_k) - \mathbf{x}^\tau(t_{k+1}; \mathbf{x}_k, t_k)\| = \mathcal{O}(\tau^p).$$

Our main theoretical result is a statement about the local error upon crossing the interface on some time interval $[t_k, t_{k+1}]$. Assuming order p schemes in the smooth regions the global error is also of order p , provided that the local error is of order p upon crossing the interface a fixed number of times. Note that we do not need to show that the local error on crossing is order $p+1$ due to the assumption of finitely many crossings on I .

4.1. An elementary lemma on quadratic inequalities. We begin with an elementary lemma about nonnegative solutions to a quadratic inequality that we will utilize for our main theorem.

Lemma 4.2 ([26, Lemma 4.2]). *Consider the inequality $0 \leq ax^2 - bx + c$ for $x \in [0, \infty)$, where a, b are fixed positive constants and c as a positive parameter. Then for $c < \frac{b^2}{4a}$, both roots of the quadratic polynomial are real and the nonnegative solutions to the inequality belong to the intervals,*

$$x \in [0, r_-(a, b, c)] \cup [r_+(a, b, c), \infty), \text{ where } r_\pm(a, b, c) := \frac{b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Furthermore, $r_-(a, b, c) \leq \frac{c}{b} + \frac{c}{b} \sum_{n=1}^{\infty} \left(\frac{2ac}{b^2} \right)^n = \frac{c}{b} + \mathcal{O}(c^2)$.

4.2. Main estimates. Next we prove two lemmas for estimating $|t - t^*|$ and $|t - \hat{t}^*|$ (which we will call times from crossing) during a single time step in which an interface crossing occurs. We also prove that a *discrete* transversality condition holds, which we utilize to prove the lemma estimating $|t - \hat{t}^*|$. The discrete transversality condition and both estimates are given in terms of various constants of the problem.

Lemma 4.3 ([26, Lemma 4.3]). *Let $g \in C^2(\bar{U} \rightarrow \mathbb{R})$ and assume the continuous transversality condition (2) is satisfied. Then for all $t \in [t_k, t_{k+1}]$, the following estimate holds:*

$$|t - t^*| \leq \frac{M(t - t^*)^2 + L_g \|\mathbf{x}(t; \mathbf{x}_k, t_k) - \mathbf{x}(t^*; \mathbf{x}_k, t_k)\|}{\alpha_S^2}, \quad (9)$$

where L_g is the uniform Lipschitz constant of g and

$$M := \frac{1}{2} \max \left\{ \sup_{t \in [t_k, t^*)} |\dot{\mathbf{x}}(t) \cdot H_g(\mathbf{x}(t)) \dot{\mathbf{x}}(t) + \nabla g(\mathbf{x}(t)) \cdot \ddot{\mathbf{x}}(t)|, \right. \\ \left. \sup_{t \in (t^*, t_{k+1}]} |\dot{\mathbf{x}}(t) \cdot H_g(\mathbf{x}(t)) \dot{\mathbf{x}}(t) + \nabla g(\mathbf{x}(t)) \cdot \ddot{\mathbf{x}}(t)| \right\}, \quad (10)$$

where H_g is the Hessian matrix of g .

Before presenting a discrete version of Lemma 4.3, we need to first show a result on the *discrete transversality condition* for the discrete vector fields $\mathbf{f}_\pm^\tau$ in relation to ∇g . The idea is based on the consistency of the discrete vector fields and the continuous transversality condition.

Lemma 4.4 ([26, Lemma 4.4]). *Let $g \in C^2(\overline{U} \rightarrow \mathbb{R})$. Suppose the discrete vector fields $\mathbf{f}_\pm^\tau$ are locally consistent of order p to the respective vector fields $\mathbf{f}_\pm$ on $I \times (U_\pm \cup S)$; that is for $t \in I, \mathbf{x} \in U_\pm \cup S$, there exists positive constants τ_0, C_1, C_2, C_3 so that for all $0 < \tau \leq \tau_0$ and if $a, b \in B_{C_1\tau_0}(t)$ and $\mathbf{y}, \mathbf{z} \in B_{C_2\tau_0^p}(\mathbf{x})$, then*

$$\|\mathbf{f}_\pm(t, \mathbf{x}) - \mathbf{f}_\pm^\tau(a, \mathbf{y}, b, \mathbf{z})\| \leq C_3\tau^p. \quad (11)$$

Then for some positive constant τ_1 , for $0 < \tau \leq \tau_1$, if $t, s \in B_{C_1\tau_1}(t^)$ and $\mathbf{x}, \mathbf{y}, \mathbf{z} \in B_{C_2\tau_1^p}(\mathbf{x}^*)$, there exist a nonzero constant $\hat{\alpha}_S \in \mathbb{R}$ so that the following discrete transversality condition holds:*

$$0 < \hat{\alpha}_S^2 \leq \nabla g(\mathbf{x}) \cdot \mathbf{f}_\pm^\tau(t, \mathbf{y}, s, \mathbf{z}) \quad (12)$$

By analogy with the continuous case, we will refer to the constant $\hat{\alpha}_S$ appearing in the above lemma as the *discrete transversality constant*.

Lemma 4.5 ([26, Lemma 4.5]). *Let $g \in C^2(\overline{U} \rightarrow \mathbb{R})$. For all $t \in [t_k, t_{k+1}]$, the following discrete analogue of the continuous estimate (9) holds for $\tau \leq \tau_0$:*

$$|t - \hat{t}^*| \leq \frac{\hat{M}(t - \hat{t}^*)^2 + L_g \|\mathbf{x}^\tau(t; \mathbf{x}_k, t_k) - \mathbf{x}^\tau(\hat{t}^*; \mathbf{x}_k, t_k)\|}{\hat{\alpha}_S^2} \quad (13)$$

Here $\hat{\alpha}_S$ is the discrete transversality constant and τ_0 is an upper bound, both defined in Lemma 4.4. L_g is the Lipschitz constant of g , and

$$\hat{M} := \frac{1}{2} \max(\hat{M}_-, \hat{M}_+), \quad (14)$$

where

$$\begin{aligned} \hat{M}_- &:= \sup_{\substack{t \in [t_k, \hat{t}^*] \\ s \in [0, 1]}} \mathbf{f}_-^\tau(t, \mathbf{x}^\tau(t), \hat{t}^*, \hat{\mathbf{x}}^*) \cdot H_g(\mathbf{x}^\tau(t) + s(\hat{\mathbf{x}}^* - \mathbf{x}^\tau(t))) \mathbf{f}_-^\tau(t, \mathbf{x}^\tau(t), \hat{t}^*, \hat{\mathbf{x}}^*), \\ \hat{M}_+ &:= \sup_{\substack{t \in [\hat{t}^*, t_{k+1}] \\ s \in [0, 1]}} \mathbf{f}_+^\tau(\hat{t}^*, \hat{\mathbf{x}}^*, t, \mathbf{x}^\tau(t)) \cdot H_g(\hat{\mathbf{x}}^* + s(\mathbf{x}^\tau(t) - \hat{\mathbf{x}}^*)) \mathbf{f}_+^\tau(\hat{t}^*, \hat{\mathbf{x}}^*, t, \mathbf{x}^\tau(t)). \end{aligned}$$

Lastly, we need one final lemma to prove the main theorem relating the spatial transition points of the piecewise smooth and discrete trajectories. This is the only instance where we require the relatively strong hypothesis of having $d - 1$ conserved quantities. However in doing so, we can guarantee that the transition time can be computed accurately, in contrast to Mannshardt's original hypothesis of requiring this a priori.

Lemma 4.6 ([26, Lemma 4.6]). *Let $\mathbf{f}_-^\tau$ be the conservative discretization of (6) which preserves $d - 1$ time-independent conserved quantities $\psi_- : U_- \cup S \subset \mathbb{R}^d \rightarrow \mathbb{R}^{d-1}$. Let $\mathbf{x}^* = \mathbf{x}(\hat{t}^*; \mathbf{x}_k, t_k)$ and $\hat{\mathbf{x}}^* = \mathbf{x}^\tau(\hat{t}^*; \mathbf{x}_k, t_k)$ such that $g(\mathbf{x}^*) = 0 = g(\hat{\mathbf{x}}^*)$ with $t^*, \hat{t}^* \in [t_k, t_{k+1}]$. Then for sufficiently small τ ,*

$$\mathbf{x}^* = \hat{\mathbf{x}}^*. \quad (15)$$

4.3. Proof of theorem 4.1.

Proof. We are now in position to prove our main result:

$$\|\mathbf{x}(t_{k+1}; \mathbf{x}_k, t_k) - \mathbf{x}^\tau(t_{k+1}; \mathbf{x}_k, t_k)\| = \mathcal{O}(\tau^p)$$

Since $\mathbf{x}(t^*; \mathbf{x}_k, t_k) = \mathbf{x}^*$, we can change the starting point of the piecewise smooth trajectory to be at $(\mathbf{x}^*, t^*)$, and so $\mathbf{x}(t_{k+1}; \mathbf{x}_k, t_k) = \mathbf{x}(t_{k+1}; \mathbf{x}^*, t^*)$. Furthermore, due to the inexactness of solving the nonlinear equations (6) and (7), we denote $(\hat{\mathbf{x}}, \hat{t})$ as a numerical approximation to the roots $(\hat{\mathbf{x}}^*, \hat{t}^*)$. As a consequence, $\mathbf{x}^\tau(\hat{t}; \mathbf{x}_k, t_k) = \hat{\mathbf{x}}$ by Algorithm 1 and we can change the starting point of the discrete trajectory to be at $(\hat{\mathbf{x}}, \hat{t})$. Thus, $\mathbf{x}^\tau(t_{k+1}; \mathbf{x}_k, t_k) = \mathbf{x}^\tau(t_{k+1}; \hat{\mathbf{x}}, \hat{t})$ and we can rewrite

$$\|\mathbf{x}(t_{k+1}; \mathbf{x}_k, t_k) - \mathbf{x}^\tau(t_{k+1}; \mathbf{x}_k, t_k)\| = \|\mathbf{x}(t_{k+1}; \mathbf{x}^*, t^*) - \mathbf{x}^\tau(t_{k+1}; \hat{\mathbf{x}}, \hat{t})\|. \quad (16)$$

Furthermore, we point out an important geometric aspect of our error analysis. By our hypothesis of Theorem 4.1, the conservative schemes on each part of the phase space preserves the $d-1$ conserved quantities. Thus, by Lemma 4.6, we have that $\|\mathbf{x}^* - \hat{\mathbf{x}}^*\| < \epsilon$. Now to prove the main result, we separate into three cases: $t^* = \hat{t}^*$, $t^* > \hat{t}^*$ and $t^* < \hat{t}^*$. In each of these cases, we need statements regarding the monotonicity of $g(\mathbf{x}(t))$ and $g(\mathbf{x}^\tau(t))$ near t^* and $\hat{t}^*$ for sufficiently small t . The Lemmas from [26, Appendix B] provide these statements and their proofs. We now proceed to prove each of the three cases.

Case $t^* = \hat{t}^*$. By equation (16), we have by triangle inequality

$$\begin{aligned} \|\mathbf{x}(t_{k+1}; \mathbf{x}^*, t^*) - \mathbf{x}^\tau(t_{k+1}; \hat{\mathbf{x}}, \hat{t})\| &\leq \underbrace{\|\mathbf{x}(t_{k+1}; \mathbf{x}^*, t^*) - \mathbf{x}^\tau(t_{k+1}; \mathbf{x}^*, t^*)\|}_{(A.1)} \\ &\quad + \underbrace{\|\mathbf{x}^\tau(t_{k+1}; \mathbf{x}^*, t^*) - \mathbf{x}^\tau(t_{k+1}; \hat{\mathbf{x}}, \hat{t})\|}_{(A.2)} \end{aligned}$$

We first deal with term (A.1). For sufficiently small τ , $g(\mathbf{x}(t_{k+1}; \mathbf{x}^*, t^*)) > 0$ by [26, Lemma B.1], which implies that $\mathbf{x}(t_{k+1}; \mathbf{x}^*, t^*)$ lies in U_+ . Similarly, for sufficiently small τ , $g(\mathbf{x}^\tau(t_{k+1}; \mathbf{x}^*, t^*)) > 0$ by [26, Lemma B.3], which implies that $\mathbf{x}^\tau(t_{k+1}; \mathbf{x}^*, t^*)$ lies in U_+ . Thus, the term (A.1) is $\mathcal{O}(\tau^p)$ by the convergence order of the scheme in the smooth region U_+ . For term (A.2), Lipschitz continuity of initial data with Lipschitz constant $\hat{L}$ implies that

$$\|\mathbf{x}^\tau(t_{k+1}; \mathbf{x}^*, t^*) - \mathbf{x}^\tau(t_{k+1}; \hat{\mathbf{x}}, \hat{t})\| \leq \hat{L} (\|\mathbf{x}^* - \hat{\mathbf{x}}\| + |t^* - \hat{t}|).$$

Because $\mathbf{x}^* = \hat{\mathbf{x}}^*$ and $t^* = \hat{t}^*$ by Lemma 4.6, both of these terms are machine precision. Thus, the main result in the case that $t^* = \hat{t}^*$ is proven.

Case $t^* > \hat{t}^*$. Again, by equation (16), we have by triangle inequality

$$\begin{aligned} \|\mathbf{x}(t_{k+1}; \mathbf{x}^*, t^*) - \mathbf{x}^\tau(t_{k+1}; \hat{\mathbf{x}}, \hat{t})\| &\leq \underbrace{\|\mathbf{x}(t_{k+1}; \mathbf{x}^*, t^*) - \mathbf{x}^\tau(t_{k+1}; \mathbf{x}^*, t^*)\|}_{(B.1)} \\ &\quad + \underbrace{\|\mathbf{x}^\tau(t_{k+1}; \mathbf{x}^*, t^*) - \mathbf{x}^\tau(t_{k+1}; \hat{\mathbf{x}}^*, \hat{t}^*)\|}_{(B.2)} \\ &\quad + \underbrace{\|\mathbf{x}^\tau(t_{k+1}; \hat{\mathbf{x}}^*, \hat{t}^*) - \mathbf{x}^\tau(t_{k+1}; \hat{\mathbf{x}}, \hat{t})\|}_{(B.3)}. \end{aligned}$$

We now show each of these terms is $\mathcal{O}(\tau^p)$, which would prove the result. For term (B.1), $\mathbf{x}(t_{k+1}; \mathbf{x}^*, t^*)$ lies in U_+ , since $g(\mathbf{x}(t_{k+1}; \mathbf{x}^*, t^*)) > 0$ for sufficiently small τ ,

by [26, Lemma B.1]. Moreover, $\mathbf{x}^\tau(t_{k+1}; \mathbf{x}^*, t^*)$ lies in U_+ , since $g(\mathbf{x}^\tau(t_{k+1}; \mathbf{x}^*, t^*)) > 0$ for sufficiently small τ , by [26, Lemma B.3]. The discrete and piecewise smooth solutions used in this term start on S and lie entirely in U_+ except at the start and so we can conclude that term (B.1) above is $\mathcal{O}(\tau^p)$.

For term (B.2), since the discrete solution is assumed uniformly Lipschitz in the initial data, with Lipschitz constant $\hat{L}$,

$$\|\mathbf{x}^\tau(t_{k+1}; \mathbf{x}^*, t^*) - \mathbf{x}^\tau(t_{k+1}; \hat{\mathbf{x}}^*, \hat{t}^*)\| \leq \hat{L}(\|\mathbf{x}^* - \hat{\mathbf{x}}^*\| + |t^* - \hat{t}^*|).$$

By equation (15), this difference reduces to estimating the time error $|t^* - \hat{t}^*|$ which is discussed below.

For term (B.3), we use the same Lipschitz continuity argument as in (B.2) and the fact that $(\hat{\mathbf{x}}, \hat{t})$ can be made to converge to $(\hat{\mathbf{x}}^*, \hat{t}^*)$ to an arbitrary precision ϵ .

Now, it remains to show that $|t^* - \hat{t}^*| = \mathcal{O}(\tau^p)$. To do so, we use Lemma 4.3, with $t = \hat{t}^*$

$$|\hat{t}^* - t^*| \leq \frac{M(\hat{t}^* - t^*)^2 + L_g \|\mathbf{x}(\hat{t}^*; \mathbf{x}_k, t_k) - \mathbf{x}(t^*; \mathbf{x}_k, t_k)\|}{\alpha_S^2},$$

where L_g is the uniform Lipschitz constant of g and M is as defined in the lemma. Equivalently,

$$0 \leq M(\hat{t}^* - t^*)^2 - \alpha_S^2 |\hat{t}^* - t^*| + L_g \|\mathbf{x}(\hat{t}^*; \mathbf{x}_k, t_k) - \mathbf{x}(t^*; \mathbf{x}_k, t_k)\|. \quad (17)$$

To estimate the norm term in (17) note that

$$\begin{aligned} \|\mathbf{x}(t^*; \mathbf{x}_k, t_k) - \mathbf{x}(\hat{t}^*; \mathbf{x}_k, t_k)\| &\leq \underbrace{\|\mathbf{x}(\hat{t}^*; \mathbf{x}_k, t_k) - \mathbf{x}^\tau(\hat{t}^*; \mathbf{x}_k, t_k)\|}_{(B.4)} \\ &\quad \underbrace{\|\mathbf{x}^\tau(\hat{t}^*; \mathbf{x}_k, t_k) - \mathbf{x}(t^*; \mathbf{x}_k, t_k)\|}_{(B.5)}. \end{aligned}$$

For term (B.4), by [26, Lemma B.2] with $t = \hat{t}^*$ and for sufficiently small τ , $g(\mathbf{x}(\hat{t}^*)) < 0$ since $t^* < \hat{t}^*$. So $\mathbf{x}(\hat{t}^*) \in U_-$. Moreover, since $g(\hat{\mathbf{x}}^*) = 0$, $\hat{\mathbf{x}}^* \in S$. Thus, both the piecewise smooth and discrete trajectories lie in $U_- \cup S$, which implies that the term (B.4) is $\mathcal{O}(\tau^p)$. For term (B.5) note that since $\mathbf{x}^\tau(\hat{t}^*) = \hat{\mathbf{x}}^*$, and $\mathbf{x}(t^*) = \mathbf{x}^*$, term (B.5) can be made as small as desired, for example less than ϵ . Thus equation (17) reduces to

$$0 \leq M|t^* - \hat{t}^*|^2 - \alpha_S^2 |t^* - \hat{t}^*| + L_g(C\tau^p + \epsilon),$$

where C is a constant implicit in $\mathcal{O}(\tau^p)$. Then since ϵ can be made as small as desired, we can absorb it into the τ^p term. Note also that $|t^* - \hat{t}^*| \leq \tau$, since both times are contained in the interval $[t_k, t_{k+1}]$. Thus, we can apply Lemma 4.2 and ensure that for sufficiently small τ , $|t^* - \hat{t}^*|$ is in the lower interval of Lemma 4.2. This implies that $|t^* - \hat{t}^*|$ is $\mathcal{O}(\tau^p)$.

Finally, the proof for the case of $t^* < \hat{t}^*$ is similar. For brevity, we refer the reader interested in the remaining part of the proof to [26, Section 4.3]. $\square$

5. Numerical results. We now showcase numerical experiments for our conservative transition scheme on two PWS systems: a two dimensional system with a cubic polynomial conserved quantity and a three species Lotka-Volterra system with a logarithmic conserved quantity.

For both systems, we computed the error before and after crossing an interface with discontinuities in their parameters. We examined the effect of adding error artificially to the computed time of intersection. allowing for a demonstration of

Theorem 4.1. Specifically, if the error introduced into the intersection time value is of an order larger than the global truncation error of the method, then the order of the method deteriorates after crossing the interface. On the other hand, if the error is smaller then it does not, as we have proved in the case of codimension one number of conserved quantites.

5.1. Elliptic curve system. Our first example is the elliptic curve example used in [25] to illustrate long-term stability of conservative integrators over traditional and symplectic integrators. The PWS is defined as $\dot{x}(t) = 2y(t)$ and $\dot{y}(t) = 3x^2(t) + a_{\pm}$ with the difference in the parameter $a_{\pm}$ introducing the discontinuity in the system. The conserved quantities for this system are $\psi_{\pm}(x, y) = y^2 - x^3 - a_{\pm}x$, which defines an elliptic curve motivating the name of the system. We emphasize here that the conserved quantities are cubic polynomials and would not be preserved in general by RK methods [16].

The phase space is divided into two regions by a circular interface of radius 1 centered at the origin with the parameter value $a_- = -3$ inside the circle and $a_+ = -2$ outside. Thus $U = \mathbb{R}^2$, $g(x, y) := x^2 + y^2 - 1$ and the switching surface S is the circle $g(x, y) = 0$, with the two parts of the phase space given by $U_{\pm} = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \gtrless 1\}$.

Using the DMM discretization of this system from [25], the discrete PWS vector fields are given by $\mathbf{f}_{\pm}^{\tau} = (y_k + y_{k+1}, x_k^2 + x_k x_{k+1} + x_{k+1}^2 + a_{\pm})$ with some trajectories shown in Figure 1a. Note that $\mathbf{f}_{\pm}^{\tau}$ is also symmetric⁵ and as expected, it was empirically observed in Figure 2 to have global truncation error of $\mathcal{O}(\tau^2)$, where the error was computed using a reference solution computed by a 4-th order Runge-Kutta integrator using a fixed time step size of $\tau = 1.6 \times 10^{-5}$.

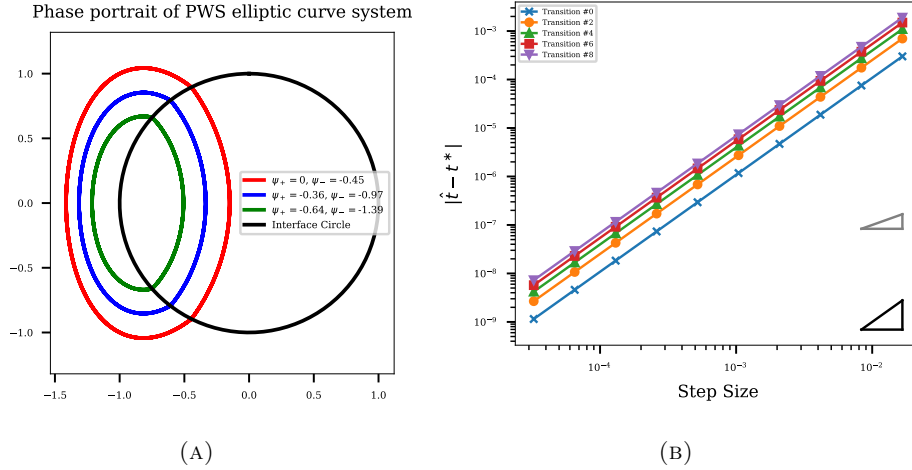


FIGURE 1. (A) PWS trajectories for the elliptic curve example with the interface being a circle of radius 1 centered at the origin. (B) Convergence of the error between the exact transition time t^* and the approximate transition time $\hat{t}$ for the PWS elliptic curve system.

⁵It is known that a symmetric one step method has even order of accuracy. See [12] for more details.

Error in $\|\mathbf{x}_k - \mathbf{x}(t_k)\|$ resulting from perturbation of $O(\tau^p)$ in $\hat{t}$

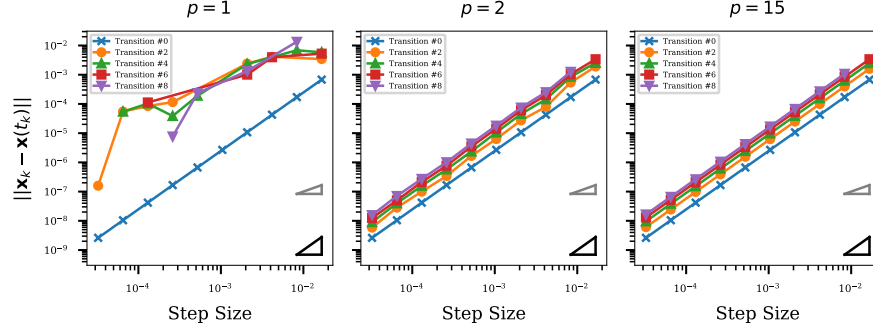


FIGURE 2. Comparison of error in the solution of the PWS elliptic curve system with different perturbation added to $\hat{t}$ each time the interface is crossed.

Both Figure 1b and Figure 2 show the behavior of error in the PWS system before and after crossing the interface when a perturbation is introduced into the computation of $\hat{t}$. The size of the perturbation added is of order $O(\tau^p)$ for $p = 1, 2, 15$. While the $p = 1$ perturbation is larger than the truncation error of the method, the $p = 2$ perturbation is of the same order and the $p = 15$ perturbation introduces effectively no error to $\hat{t}$. Specifically, the first plot in Figure 2 shows that if the intersection time is not computed accurately enough, the accuracy degrades to $O(\tau)$ as originally predicted by Mannshardt [17]. In contrast, the last plot in Figure 2 confirms that when negligible perturbation is added to $\hat{t}$, the order of accuracy stays the same, even after the interface has been crossed multiple times.

Figure 3 shows the ability of the DMM transition scheme to accurately preserve the conserved quantities on the two sides of the interface even after multiple transitions. As expected, the error in the conserved quantity is at machine precision for DMM, whereas the RK2 integrator shows an error of about 10^{-7} with a drift in the conserved quantity over time.

5.2. Three species Lotka-Volterra system. Our second example is a three species Lotka-Volterra system studied in [22, 23] given by, $\dot{\mathbf{x}}(t) = \text{diag}(\mathbf{x}(t))A_{\pm}(\mathbf{x}(t) - \boldsymbol{\xi})$, where $\mathbf{x}(t) = (x_1(t), x_2(t), x_3(t))^T \in \mathbb{R}_+^3$ are the relative population of three species at time t , $A_{\pm}$ is a constant 3×3 interaction matrix and $\boldsymbol{\xi} \in \mathbb{R}_+^3$ is a fixed point. If $A_{\pm}$ is skew-symmetric, then [20] showed a conserved quantity of the form $\psi(\mathbf{x}) = \boldsymbol{\xi}^T \log(\mathbf{x}) - \mathbf{1}^T \mathbf{x}$. For the interface, we chose $g(\mathbf{x}) = x + y + z - 1$ on $U = \mathbb{R}_+^3$ and $\boldsymbol{\xi} = (1, 1, 1)$. Thus the two parts of the phase space are $U_{\pm} = \{\mathbf{x} \in \mathbb{R}_+^3 \mid g(\mathbf{x}) \gtrless 0\}$. Using the DMM discretization of this system from [23], the discrete PWS vector fields from DMM is given by $\mathbf{f}_{\pm}^T = \bar{\mathbf{x}} \circ A_{\pm}(\bar{\mathbf{x}} \circ (1 - \boldsymbol{\xi} \circ \frac{\Delta \log(\mathbf{x}_k)}{\Delta \mathbf{x}_k}))$, where $\circ$ refers to the element-wise product, $\bar{\mathbf{x}} = (\mathbf{x}_k + \mathbf{x}_{k-1})/2$ and $\frac{\Delta \log(\mathbf{x}_k)}{\Delta \mathbf{x}_k} = \frac{1}{\mathbf{x}_k} \circ \log(\frac{\mathbf{x}_{k+1}}{\mathbf{x}_k}) / (\frac{\mathbf{x}_{k+1}}{\mathbf{x}_k} - 1)$ where all operations are performed element-wise. The piecewise skew symmetric matrix $A_{\pm}$ has entries $A_{1,2} = \frac{1}{15}$, $A_{1,3} = -\frac{14}{15}$, and $A_{2,3} = 1$ for $g > 0$ and $A_{1,2} = \frac{14}{15}$, $A_{1,3} = -\frac{31}{15}$, and $A_{2,3} = 2$ for $g < 0$.

Since this example has only one conserved quantity in three dimensions, it does not satisfy the hypothesis of our main result of Theorem 4.1. On the other hand,

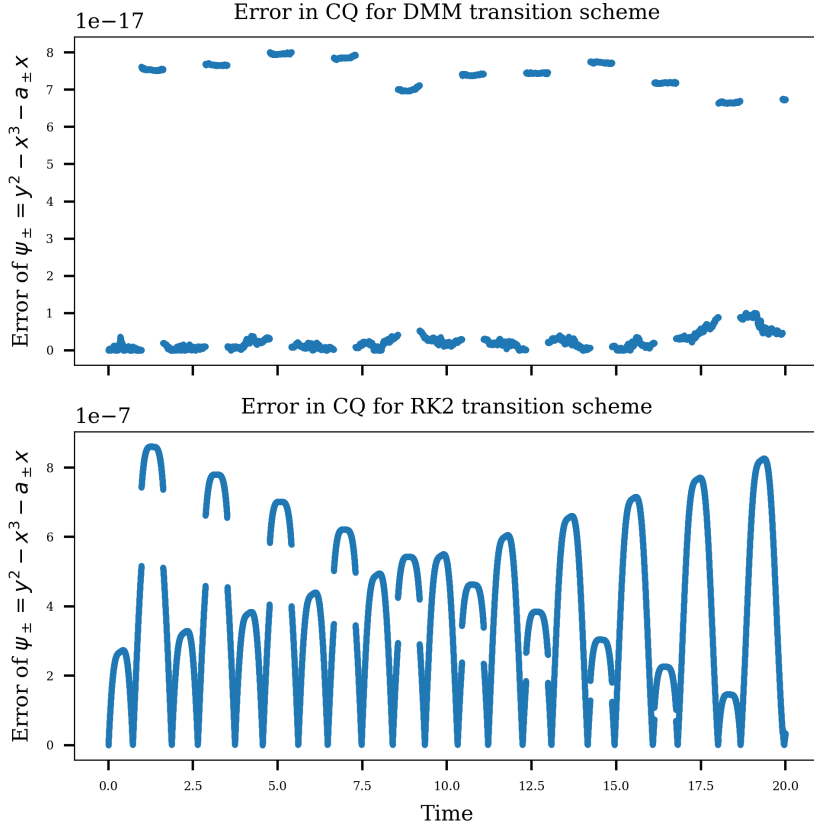


FIGURE 3. Comparison of the error in conserved quantity versus time between our DMM transition scheme and the RK2 transition scheme for the PWS elliptic curve system. An initial condition of $[-1, -1]$ and a step size 10^{-3} was used for both methods.

for sufficiently small time step size, the order preservation across transitions would still hold by Mannshardt's original results. However, as the numerical results show, spurious transitions can occur for transition schemes using the RK2 integrator, in contrast to using the conservative DMM integrator. This is because the DMM scheme enables its solutions to remain close to the conserved quantity up to machine precision, yielding long-term solutions with more accurate transition points.

Specifically, comparison of trajectories of Figure 4a shows that the RK2 solution drifts very far off from the RK4 and DMM solution, due to its spurious transitions which are demonstrated by Figure 4a.

Figures 4b and 5 are the error plots for the three species Lotka-Volterra system analogous to the error plots in Figures 1b and 2 for the elliptic curve system. As noted above, our order preservation result does not apply here but Mannshardt's original result still guarantees order preservation.

Since no transition occurs in the reference solution for this initial condition, the value of ψ remains constant over the entire trajectory. Thus, we can compare the error in ψ across the trajectories of RK2, RK4 and DMM, as shown in Figure 6.

Moreover, Figure 7 shows the RK2 trajectories makes spurious transitions due to drifting away from the conserved quantity, leading to inaccurate interface crossing.

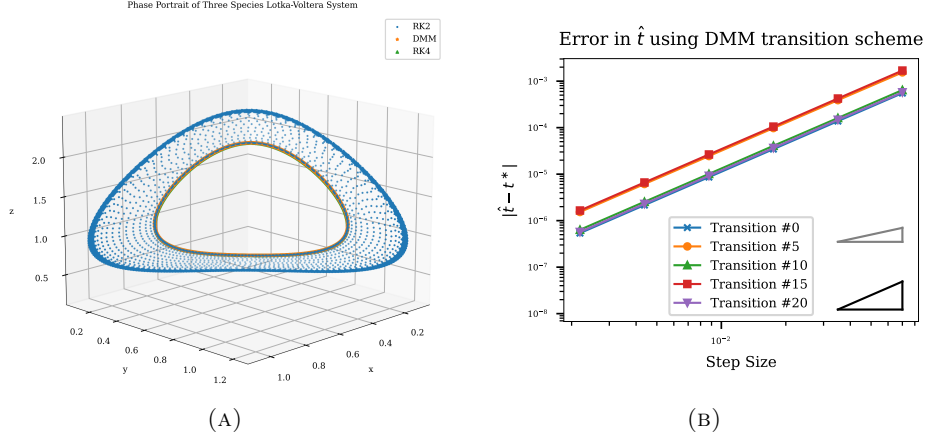


FIGURE 4. (A) An initial condition of $[0.612, 0.0137, 0.629]$ was used with a final time $T = 700$ and step size $\tau = 0.07$. Here DMM transition scheme agrees with the RK4 transition scheme, but the RK2 transition scheme has drifted far away due to spurious transitions. (B) Convergence of the error between the exact transition time t^* and the approximate transition time $\hat{t}$ for the PWS three species Lotka-Volterra system.

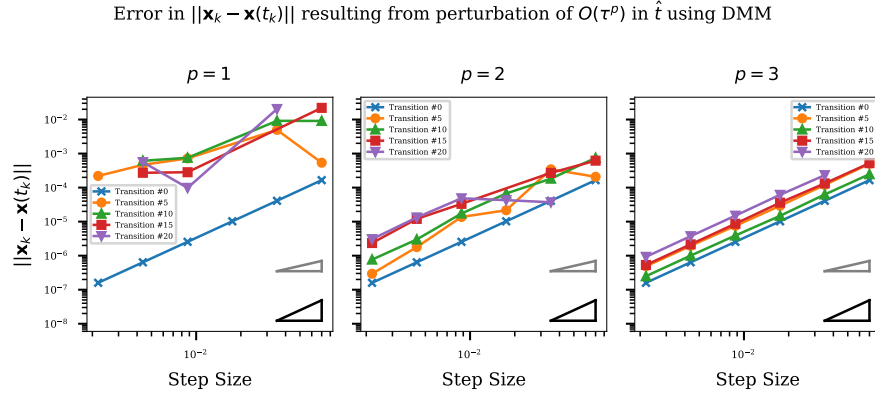


FIGURE 5. Comparison of error in the solution of the PWS three species Lotka-Volterra system with different perturbation added to $\hat{t}$ each time the interface is crossed.

As another numerical experiment, using the initial condition of $[0.612, 0.1345, 0.629]$, Figure 9 demonstrates that all three methods makes transitions across the interface initially. However, RK2 still makes spurious crossings of the interface at a later time. From Figure 8, we observed that the jumps in the error in ψ for RK2 correspond to the spurious crossing as shown in Figure 9.

Thus, for more complex conservative piecewise smooth systems, trajectories that drift away from a conserved quantity can lead to dramatic changes to interface crossings resulting in spurious transitions, especially over long-term integration. In contrast, conservative transition schemes may result in more reliable solutions at interface crossings for such piecewise smooth systems.

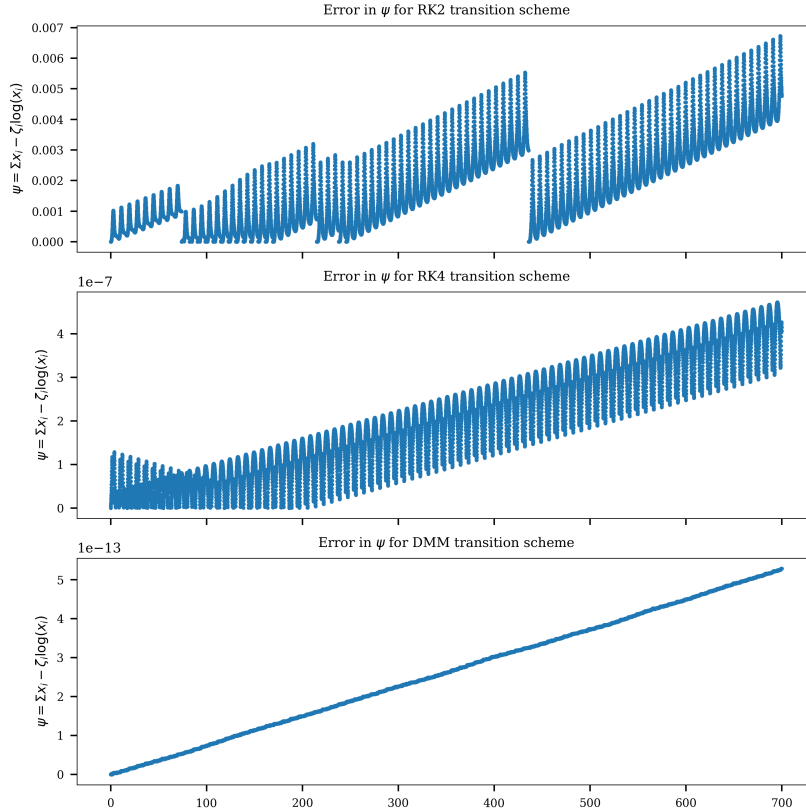


FIGURE 6. Comparison of the error in conserved quantity versus time, as computed by RK2 (top), RK4 (middle) and our DMM transition scheme (bottom) for the PWS three species Lotka-Volterra system.

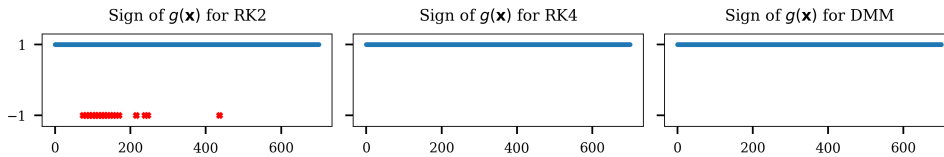


FIGURE 7. The sign of the switching function g for the three methods are plotted versus time, corresponding to Figure 6. In this case, the solution computed using RK2 makes spurious transitions (denoted by red "x") while the RK4 and DMM solutions do not.

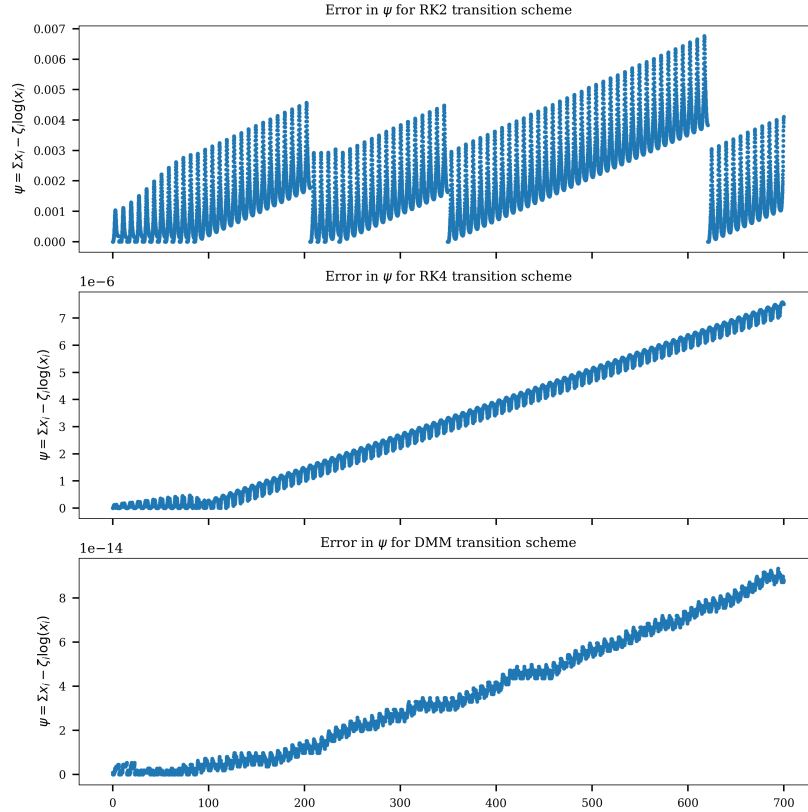


FIGURE 8. Comparison of error in conserved quantity versus time, as computed by RK2 (top), RK4 (middle) and our DMM transition scheme (bottom) for the PWS three species Lotka-Volterra system when the solution makes several transitions.

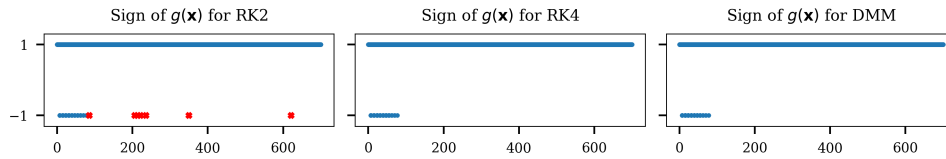


FIGURE 9. The sign of the switching function g for the three methods are plotted versus time, corresponding to Figure 8. All three methods make initial transitions and then RK2 makes further spurious transitions (denoted by red “x”).

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